Biological molecules

Carbohydrates are made of hydrogen, carbon, and oxygen ($C_6H_{12}O_6$). They are divided to three groups:

1- Monosaccharides : Glucose, galactose, Fructose

* Sweet and soluble in water

2- Disaccharides : Two monosaccharides joined together by an Oxygen.

- Sucrose (table sugar): Glucose + Fructose
- lactose (Milk sugar): Glucose + Galactose
- Maltose (Malt sugar): Glucose + Glucose

* Sweet and soluble in water

3- Polysaccharides : Many monosaccharides joined together

- Startch
- Glycogen
- Cellulose

* Not sweet and insoluble in water

Uses of carbohydrates

* For supply of instant energy since they are easily digested.
* Release energy in respiration

→ Glucose is transported in the plasma of the blood
→ Animals store glucose as glycogen in the liver

→ Plants transport the sugar sucrose to cells
→ Sucrose is converted to glucose for respiration
→ Plants store glucose in the form of starch

Fats: Made of carbon, hydrogen and oxygen, which make three fatty acids and glycerol

Uses of Fats:

1- Making cell membrane

2- Store and release energy. (Fats are stored in the adipase tissue under the skin)

3- Making cholestrol, which is used in cell membrane

4- Making steriod hormones: Testosterone, oestrogen

5- For insulation and to prevent heat loss

Proteins: Made of smaller units called amino acids., which are made of hydrogen, carbon, oxygen, nitrogen, sometimes sulfur.

Uses of protein

1- Growth and repair

2- Making antibodies

3- Making enzymes

4- Making hemoglobin which transports oxygen around the body

5- To make DNA

6- For energy only when carbohydrates and fats are completely used up

* 80% of the body is made up of water

Uses of Water

1- As a solvent

2- As a medium for metabolic reactions

3- Makes up most of the plasma

4- To cool the body when lost as sweat

5- Helps in getting rid of wastes like urea

A DNA molecule is made up of nucleotides, formed into two strands, joined to a phosphate group and an organic base. The sugar in DNA is deoxyribose and the base is either: A, T, C, G

Testing for food

1) Testing for reducing sugars: Benedict solution:

Add the Benedict solution to the unknown solution, and heat in a water bath, detect color change.

Blue → Green → Yellow → Orange → Red

2) Testing for vitamin C: DCPIP

The solution will change from blue to colorless

3) Testing for startch: Iodine solution

The solution changes color from Reddish-Brown to blue-black

4) Testing for protein: Biuret

Changes from Blue to Purple

5) Testing for Fat: Ethanol

Changes from colorless to cloudy/milky white